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CLASS: AL1/A14
SUBJECT: ESSENTIALS OF DATA SCIENCE (EDS)

practical 4 :

4.1.1

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.
code and output :

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
8
9 # Grouping by even and odd numbers and calculating the mean
10 grouped = series.groupby(series % 2 == 0).mean()
11
12 # Display the mean of even and odd numbers with labels
13 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
14 print("Mean of even and odd numbers:")
15 print(grouped)
```

Average time: 0.010 s (9.67 ms) | Maximum time: 0.025 s (25.00 ms) | 3 out of 3 shown test case(s) passed | 3 out of 3 hidden test case(s) passed

Test case 1 (25 ms) [Debug]

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd ... 5.0	Odd ... 5.0
Even ... 6.0	Even ... 6.0
dtype: float64	dtype: float64

Terminal | Test cases

4.1.2

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

- dataframeOps.py

▼

```
#import pandas as pd
#df = pd.DataFrame(data)
```

```
## Display the original DataFrame
#print("Original DataFrame:")
#print(df)
```

```
## Adding a new row
```

```
## Display the DataFrame after adding a new row
```

```
#print("After adding a row:\n",df)
```

```
## Modifying a row
```

```
## Display the DataFrame after modifying a row
```

```
#print("After modifying a row:")
```

```
#print(df)
```

```
## Deleting a row
```

```
## Display the DataFrame after deleting a row
```

```
#print("After deleting a row:")
```

```
#print(df)
```

```
## Adding a new column
```

```
## Display the DataFrame after adding a new column
```

```
#print("After adding a new column:")
```

```
#print(df)
```

```
## Modifying a column
```

```
## Display the DataFrame after modifying a column
```

```
#print("After modifying a column:")
```

```
#print(df)
```

```
## Deleting a column
```

```
## Display the DataFrame after deleting a column
```

```
#print("After deleting a column:")
```

```
#print(df)
```

```
import pandas as pd
```

```
#Provided dictionary of lists
```

```
data = {
```

```
'Name': ['Alice', 'Bob', 'Charlie'],
```

```
'Age': [25, 30, 35],
```

```
}
```

```
#Convert the dictionary to a DataFrame
```

```
df= pd.DataFrame(data)
```

```
#Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
#Adding a new row
```

```
new_name=input("New name: ")
```

```
new_age=int(input("New age: "))
```

```
new_row={'Name':new_name,'Age':new_age}
df=pd.concat([df,pd.DataFrame([new_row])],ignore_index=True)
# Display the DataFrame after adding a new row
print("After adding a row:\n",df)
```

```
#Modifying a row
modify_index=int (input("Index of row to modify: "))
new_age_mod=int(input ("New age: "))
df.loc[modify_index,"Age"]=new_age_mod
```

```
#Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)
```

```
#Deleting a row
delete_index=int(input("Index of row to delete: "))
```

Translation Failed

Reason for late submission

Please enter at least 15 characters

code :

```
import pandas as pd
```

```
#Provided dictionary of lists
```

```
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}
```

```
#Convert the dictionary to a DataFrame
```

```
df= pd.DataFrame(data)
```

```
#Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
#Adding a new row
```

```
new_name=input("New name: ")
```

```
new_age=int(input("New age: ")) new_row=
```

```
{'Name':new_name,'Age':new_age}
```

```
df=pd.concat([df,pd.DataFrame([new_row])],ignore_index=True)
```

```
# Display the DataFrame after adding a new row
```

```
print("After adding a row:\n",df)
```

```
#Modifying a row
```

```
modify_index=int (input("Index of row to modify: "))
```

```
new_age_mod=int(input ("New age: "))
```

```
df.loc[modify_index,"Age"]=new_age_mod
```

```
#Display the DataFrame after modifying a row
```

```
print("After modifying a row:")
```

```
print(df)
```

```
#Deleting a row
```

```
delete_index=int(input("Index of row to delete: "))
```

```
df=df.drop(delete_index).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row
```

```
print("After deleting a row:")
```

```
print(df)
```

```
#Adding a new column
```

```
gender_input=input("Enter genders separated by space: ")
```

```
genders=gender_input.split()
```

```
df["Gender"]= genders
```

```
#Display the DataFrame after adding a new column
```

```
print("After adding a new column:")
```

```
print(df)
```

```
#Modifying a column
```

```
df["Name"]=df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column
```

```
print("After modifying a column:")
```

```
print(df)
```

```
#Deleting a column
```

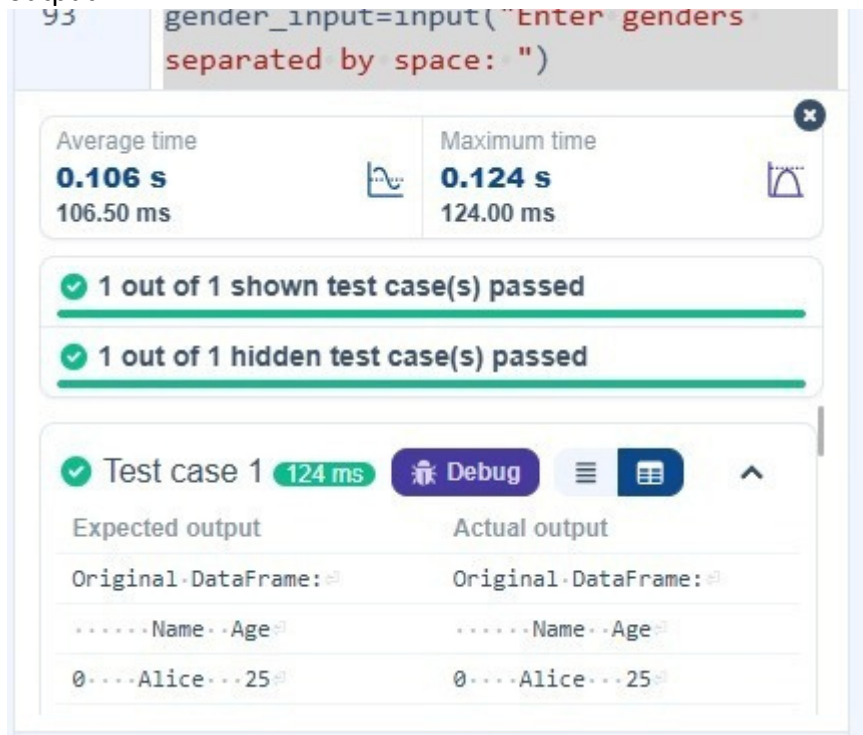
```
df=df.drop(columns=['Age'])
```

```
#Display the DataFrame after deleting a column
```

```
print("After deleting a column:")
```

```
print(df)
```


output :



4.1.3

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Code :

```
import pandas as pd
```

```
#Read the text file into a DataFrame
```

```
file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])

#Display the first five rows

print("First five rows:")

print(data.head())

#Calculate the average age (rounded to 2 decimal places)

average_age = round(data['Age'].mean(), 2)

print(f"Average age: {average_age}")

#Filter students with grade up to threshold 'B'

grade_order = {'A': 1, 'B': 2, 'C': 3, 'D': 4, 'F': 5}

threshold = 'B'

filtered_df = data[data['Grade'].map(grade_order) <= grade_order[threshold]]

print(f"Students with a grade up to {threshold}")
```

```
print(filtered_df)
```

output :

The screenshot shows a code editor with the following Python code:

```
to 2 decimal places)
average_age =
round(data['Age'].mean(), 2)
print(f"Average age: {average_age}")
```

Below the code, there is a performance summary table:

Average time	Maximum time
0.047 s 47.00 ms	0.068 s 68.00 ms

Below the table, there are test case results:

- ✓ 1 out of 1 shown test case(s) passed
- ✓ 1 out of 1 hidden test case(s) passed
- ✓ Test case 1 **68 ms**

At the bottom of the editor, there are icons for running the code and viewing the output.

4.2.1

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago 2025-
01-03,Product B,2,15,Chicago 2025-01-
03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York 2025-01-
04,Product B,5,15,Los Angeles 2025-01-
05,Product A,3,20,Chicago 2025-01-
05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code :

```
import pandas as pd
```

```
#Prompt the user for the file name
```

```
file_name = input()
```

```
#Load the data
```

```
df= pd.read_csv(file_name)
```

```
#Convert 'Date' to datetime
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
#Calculate total sales for each row
```

```
df['TotalSales'] = df['Quantity'] * df['Price']
```

```
#Extract month in YYYY-MM format
```

```
df['Month'] = df['Date'].dt.to_period('M')
```

#Group by month and sum total sales

```
monthly_sales = df.groupby('Month')['TotalSales'].sum()
```

#Find the month with highest total sales

```
best_month = monthly_sales.idxmax()
```

```
highest_sales = monthly_sales.max()
```

```
print(f"Best month: {best_month}")
```

```
print(f"Total sales: ${highest_sales:.2f}")
```

output :

The screenshot shows a coding platform interface with the following elements:

- Performance Metrics:**
 - Average time: **0.030 s** (29.67 ms)
 - Maximum time: **0.061 s** (61.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details:**
 - Test case 1 (61 ms) with a **Debug** button.
 - Expected output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
 - Actual output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
- Navigation:** A bottom bar with buttons: < Prev, Reset, Submit, Next >.

4.2.2

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample

Data:

```
Date,Product,Quantity,Price,City 2025- 01-
01,Product A,5,20,New York 2025-01-
01,Product B,3,15,Los Angeles 2025-01-
02,Product A,7,20,New York 2025-01-
02,Product C,4,30,Chicago 2025-01-
03,Product B,2,15,Chicago 2025-01-
03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York 2025-01-
04,Product B,5,15,Los Angeles 2025-01-
05,Product A,3,20,Chicago 2025-01-
05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code :

```
import pandas as pd
```

```
#Prompt the user for the file name
```

```
file_name = input()
```

#Load the CSV data

```
df= pd.read_csv(file_name)
```

#Group by Product and sum the Quantity

```
product_sales = df.groupby('Product')['Quantity'].sum()
```

#Find the product with the highest quantity sold

```
best_product = product_sales.idxmax()
```

```
highest_quantity = product_sales.max()
```

#Display the result

```
print(f"Best selling product: {best_product}")
```

```
print(f"Total quantity sold: {highest_quantity}")
```

output :

The screenshot displays a code editor with the following Python code:

```
25 # Group by Product and sum the quantity  
26 product_sales = df.groupby('Product')
```

Below the code, the execution results are shown:

Average time	Maximum time
0.022 s 22.33 ms	0.047 s 47.00 ms

Test case results:

- ✓ 1 out of 1 shown test case(s) passed
- ✓ 2 out of 2 hidden test case(s) passed
- ✓ Test case 1 47 ms

At the bottom of the editor, there are navigation buttons: < Prev, Reset, Submit, and Next >. The 'Submit' button is highlighted in green.

4.2.3

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample

Data:

```
Date,Product,Quantity,Price,City 2025- 01-
01,Product A,5,20,New York 2025-01-
01,Product B,3,15,Los Angeles 2025-01-
02,Product A,7,20,New York 2025-01-
02,Product C,4,30,Chicago 2025-01-
03,Product B,2,15,Chicago 2025-01-
03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York 2025-01-
04,Product B,5,15,Los Angeles 2025-01-
05,Product A,3,20,Chicago 2025-01-
05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd
```

```
#Prompt the user for the file name
```

```
file_name = input()
```


#Load the CSV data

```
df= pd.read_csv(file_name)
```

#Group by City and sum the Quantity

```
city_sales = df.groupby('City')['Quantity'].sum()
```

#Find the city with the highest total quantity sold

```
best_city = city_sales.idxmax()
```

#Display the result

```
print(f"City sold the most products: {best_city}")
```

output :

The screenshot displays a code editor window with a Python script and its execution results. The script, located at line 19, reads a CSV file and groups data by city to find the city with the highest total quantity sold. The output shows that Los Angeles is the city that sold the most products.

```
19 df = pd.read_csv(file_name)
```

Performance Metrics:

Metric	Value
Average time	0.021 s (21.00 ms)
Maximum time	0.043 s (43.00 ms)

Test Results:

- 1 out of 1 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1 (43 ms):

Expected output	Actual output
sales_data.csv	sales_data.csv
City sold the most products: Los Angeles	City sold the most products: Los Angeles

Navigation Buttons: < Prev, Reset, Submit, Next >

4.2.4

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Sample	Data:
Date,Product,Quantity,Price,City	2025-
01-01,Product A,5,20,New York	2025-01-
01,Product B,3,15,Los Angeles	2025-01-
02,Product A,7,20,New York	2025-01-
02,Product C,4,30,Chicago	2025-01-
03,Product B,2,15,Chicago	2025-01-
03,Product A,8,20,Los Angeles	2025-01-
04,Product C,6,30,New York	2025-01-
04,Product B,5,15,Los Angeles	2025-01-
05,Product A,3,20,Chicago	2025-01-
05,Product C,10,30,Los Angeles	

Explanation:

Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B:** Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C:** Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C:** Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd
from itertools import combinations
from collections import Counter

#Prompt user to input the file name
file_name = input()

#Read data from CSV
df= pd.read_csv(file_name)

#Initialize a list to store all product pairs
all_pairs = []

#Group by each date
for date, group in df.groupby('Date'):
    products = group['Product'].tolist()
    #Generate all unique pairs for this date
    pairs = combinations(sorted(products), 2)
    all_pairs.extend(pairs)

#Count frequency of each product pair
```

```

pair_counts = Counter(all_pairs)

#Find the maximum frequency
max_freq = max(pair_counts.values())

#Get all pairs that have the maximum frequency
most_frequent_pairs = [pair for pair, count in pair_counts.items() if count == max_freq]

#Display the result in the exact expected format
for pair in most_frequent_pairs:
    print(f"{pair[0]} and {pair[1]}: {max_freq} times")

```

output:

The screenshot shows a coding platform interface. At the top, a code editor displays the line: `max_freq = max(pair_counts.values())`. Below the editor, a performance summary box shows:

- Average time: **0.023 s** (22.67 ms)
- Maximum time: **0.048 s** (48.00 ms)

 Below this, two green checkmarks indicate:

- 1 out of 1 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

 A third section shows 'Test case 1' with a green checkmark and a green badge indicating '48 ms'. At the bottom of the interface, there are four buttons: '< Prev' (light blue), 'Reset' (yellow), 'Submit' (green), and 'Next >' (light blue).

4.2.5

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the

dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code :

```
import pandas as pd
```

```
import numpy as np
```

```
#Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
#1. Display the first 5 rows of the dataset
```

```
print(data.head())
```

```
#2. Display the last 5 rows of the dataset
```

```
print(data.tail())
```

```
#3. Get the shape of the dataset
```

```
print(data.shape)
```

```
#4. Get a summary of the dataset (info)
```

```
print(data.info())
```

```
#5. Get basic statistics of the dataset
```

```
print(data.describe())
```

```
#6. Check for missing values
```

```
print(data.isnull().sum())
```

#7. Fill missing values in the 'Age' column with the median age

```
data['Age'] = data['Age'].fillna(data['Age'].median())
```

#8. Fill missing values in the 'Embarked' column with the mode

```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

#9. Drop the 'Cabin' column due to many missing values

```
data = data.drop(columns=['Cabin'])
```

#10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

```
data['FamilySize'] = data['SibSp'] + data['Parch']
```

output :

The screenshot displays a Jupyter Notebook interface. On the left, a portion of the code editor is visible, showing the following code:

```
1, 7.25  
hayer  
310  
, fema  
8.05,  
8625  
9909,  
erg)"  
14, 1, 0
```

The main area of the notebook shows the output of a test case. At the top, performance metrics are displayed:

Average time	Maximum time
0.216 s	0.216 s
216.00 ms	216.00 ms

Below the metrics, a green checkmark indicates that the test case passed:

1 out of 1 shown test case(s) passed

The test case details are as follows:

Test case 1	216 ms	Debug	
Expected output			
Actual output			
PassengerId	Survived	PassengerId	Survived
Pclass	Fare	Pclass	Fare
Embarked	Cabin	abin	Embarked
0	1	0	1
3	7.2500	NaN	7.2500
S		NaN	S

At the bottom of the notebook, there are navigation buttons: < Prev, Reset, Submit, and Next >.

4.2.6

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd  
import numpy as np
```

#Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv')  
data['FamilySize'] = data['SibSp'] + data['Parch']
```

#1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)

```
data['IsAlone'] = (data['FamilySize'] == 0).astype(int)
```

#2. Convert 'Sex' to numeric (male: 0, female: 1)

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

#3. One-hot encode the 'Embarked' column

```
data = pd.get_dummies(data, columns=['Embarked'])
```

#4. Get the mean age of passengers

```
print(data['Age'].mean())
```

#5. Get the median fare of passengers

```
print(data['Fare'].median())
```

#6. Get the number of passengers by class

```
print(data['Pclass'].value_counts())
```

#7. Get the number of passengers by gender

```
print(data['Sex'].value_counts())
```

#8. Get the number of passengers by survival status

```
print(data['Survived'].value_counts())
```

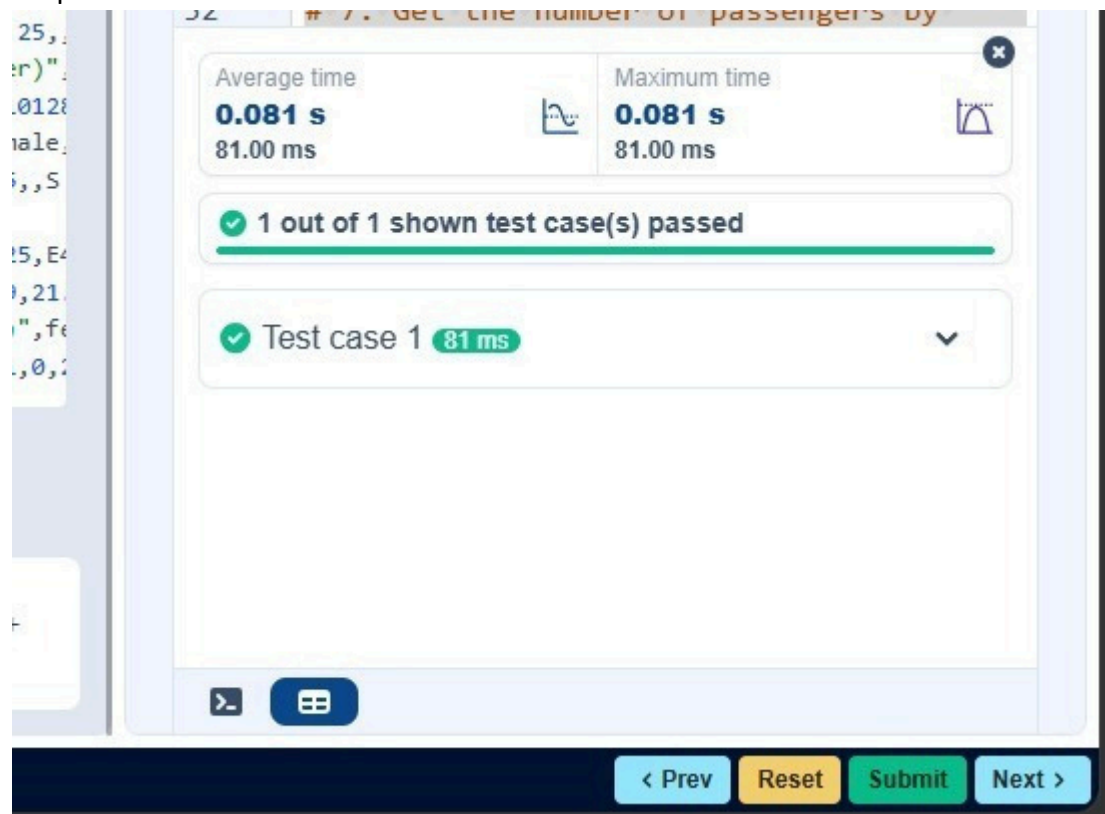
#9. Calculate the survival rate

```
print(data['Survived'].mean())
```

#10. Calculate the survival rate by gender

```
print(data.groupby('Sex')['Survived'].mean())
```

output:



The screenshot shows a coding challenge interface. On the left, a snippet of Python code is visible:

```
25, ...  
er)" ...  
.0128 ...  
ale ...  
i, S ...  
15, E ...  
1, 21 ...  
", fe ...  
.0, ...
```

The main panel displays the test results for problem #7, "Get the number of passengers by gender". At the top, it shows performance metrics: Average time 0.081 s (81.00 ms) and Maximum time 0.081 s (81.00 ms). Below this, a green bar indicates "1 out of 1 shown test case(s) passed". A dropdown menu shows "Test case 1" with a green checkmark and a timer of 81 ms. At the bottom of the panel are icons for a terminal and a settings menu.

At the very bottom of the interface, there is a navigation bar with four buttons: "< Prev", "Reset", "Submit", and "Next >".

4.2.7

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd  
import numpy as np
```

#Load the Titanic dataset

```
data = pd.read_csv('Titanic-Dataset.csv')  
data['FamilySize'] = data['SibSp'] + data['Parch']  
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)  
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

#1. Calculate the survival rate by class

```
print(data.groupby('Pclass')['Survived'].mean())
```

#2. Calculate the survival rate by embarked location

```
print(data.groupby('Embarked_S')['Survived'].mean())
```

#3. Calculate the survival rate by family size

```
print(data.groupby('FamilySize')['Survived'].mean())
```

#4. Calculate the survival rate by being alone

```
print(data.groupby('IsAlone')['Survived'].mean())
```

5. Get the average fare by class

```
print(data.groupby('Pclass')['Fare'].mean())
```

#6. Get the average age by class

```
print(data.groupby('Pclass')['Age'].mean())
```

#7. Get the average age by survival status

```
print(data.groupby('Survived')['Age'].mean())
```

#8. Get the average fare by survival status

```
print(data.groupby('Survived')['Fare'].mean())
```

#9. Get the number of survivors by class

```
print(data[data['Survived'] == 1]['Pclass'].value_counts())
```

#10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

output :

The screenshot shows a test runner interface. At the top, it displays 'Average time' and 'Maximum time' both as 0.125 s (125.00 ms). Below this, a green bar indicates '1 out of 1 shown test case(s) passed'. The test case details show 'Test case 1' with a duration of 125 ms and a 'Debug' button. A table compares 'Expected output' and 'Actual output' for 'Pclass'. The values for Pclass are 1, 2, and 3, each followed by a long decimal string. The variable name is shown as 'Name: .Survived. .dtype: .floo'. At the bottom, there are navigation buttons: '< Prev', 'Reset', 'Submit', and 'Next >'.

Expected output	Actual output
Pclass	Pclass
1...0.629630	1...0.629630
2...0.472826	2...0.472826
3...0.242363	3...0.242363
Name: .Survived. .dtype: .floo	Name: .Survived. .dtype: .floo

4.2.8

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd
import numpy as np
```

```
#Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
#1. Get the number of survivors by gender
```

```
print(data[data['Survived'] == 1]['Sex'].value_counts())
```

#2. Get the number of non-survivors by gender

```
print(data[data['Survived'] == 0]['Sex'].value_counts())
```

#3. Get the number of survivors by embarked location

```
print(data[data['Survived'] == 1]['Embarked_S'].value_counts())
```

#4. Get the number of non-survivors by embarked location

```
print(data[data['Survived'] == 0]['Embarked_S'].value_counts())
```

#5. Calculate the percentage of children (Age < 18) who survived

```
children = data[data['Age'] < 18]
```

```
print(children['Survived'].mean())
```

#6. Calculate the percentage of adults (Age >= 18) who survived

```
adults = data[data['Age'] >= 18]
```

```
print(adults['Survived'].mean())
```

#7. Get the median age of survivors

```
print(data[data['Survived'] == 1]['Age'].dropna().median())
```

#8. Get the median age of non-survivors

```
print(data[data['Survived'] == 0]['Age'].dropna().median())
```

#9. Get the median fare of survivors

```
print(data[data['Survived'] == 1]['Fare'].dropna().median())
```


10. Get the median fare of non-survivors

```
print(data[data['Survived'] == 0]['Fare'].dropna().median())
```

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`['Age'].dropna().median()`

Average time
0.082 s
82.00 ms

Maximum time
0.082 s
82.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 82 ms Debug

Expected output	Actual output
female....233	female....233
male.....109	male.....109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male.....468	male.....468
female....81	female....81

> ⌵

< Prev Reset Submit Next >

